Physical Chemistry CUME, Mar. 11th, 2016

Entropy

Q1. Entropy of mixing.

(a) Assuming a container has two chambers separated by an impermeable barrier. The left chamber has one mole of ideal gas A and the right chamber has one more of ideal gas B. Assume the barrier is now removed and the gas is allowed to mix completely. What is the entropy of mixing gas A and gas B?

(b) What is the heat of mixing?

(c) Is the mixing process reversible or nonreversible?

Q2. The third law of thermodynamics states that the entropy of a perfect crystalline at 0 K is zero. It has been hypothesized that this law is not valid for the normal ice (also called ice-Ih) because an oxygen atom in ice-Ih can form a hydrogen bond with a nearest neighbor oxygen atom in two different ways: (1) as a proton doner (**O**-H....O), or (2) as a proton acceptor (**O**...H-O).

In ice-Ih, there are 4 nearest neighbor water molecules around each water, thus each oxygen forms two donor hydrogen bonds and two acceptor bonds. There are more than one way to distribute the protons around each oxygen to form two donor and two accepter hydrogen bonds. The Pauling entropy of ice is the residual entropy due to the possibility for different hydrogen bond arrangements. Assuming all hydrogen bond arrangements with two-donor and two accepter hydrogen bonds for each water has equal energy.

(a) How many ways, W_{ind}, are there to orient the two OH bonds of each water in ice-Ih to form two donor hydrogen bonds and two accepter hydrogen bonds with its four neighbors?

(b) The proton orientations of adjacent water molecules are not independent. Assuming water_I is donating a proton to water_II, water_II cannot donate a proton to water_I at the same time. Also if water_I is adjacent to water_II, at least one of the two waters will donate a proton to the other water to form a hydrogen bond. Thus for N water molecules, the total number of ways are not W_{ind}^{N}. Please write an expression for the **entropy** of N W_B water molecules that properly consider the forbidden proton arrangements.

BONUS:

(c) In thermodynamics, particles can be classified in one of the three categories:

A. identical and distinguishable,
B. identical and indistinguishable
C. non-identical and distinguishable

For example, for a gas formed by He, all the gas molecules are identical and indistinguishable. However, if He is pressurized at low pressure and form a solid, the He atoms in the solid will be identical and distinguishable, since each He atom will have a different location.

(c1). Consider an ice-Ih sample containing N water molecules referred to as W_B, (B stands for bulk). Assuming the sample is infinitely large so that there is no water at the surface of ice. Please classify all of the W_B waters into one of the three categories.

(c2). Now consider a water molecule, W_S, (s stands for surface) at the surface of ice and remain in a solid like state. (This happens at low temperature). Please classify W_S and W_B according to the above 3 categories. (Not Ws and Wb separately, if you take a molecule of Ws and a molecule of W_B, how would you classify two molecules?)

(c3). Close to the melting temperature, it is believed that the surface of ice is actually a liquid, which is frequently referred to as the premelting layer or quasi-liquid layer. In this case, please classify Ws according to the 3 categories.